Задача №1

$$A = \begin{pmatrix} 1 & -9 \\ 7 & -60 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} 1 & -9 \\ 0 & 3 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 \\ 7 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -9 \\ 7 & -60 \end{pmatrix} \Big) (1) \cdot -7$$

$$\begin{pmatrix} 1 & -9 \\ 0 & 3 \end{pmatrix}$$

Задача №2

$$A = \begin{pmatrix} 4 & -4 & 2 & -2 \\ -4 & 5 & 1 & -2 \\ 2 & 1 & 14 & -11 \\ -2 & -2 & -11 & 19 \end{pmatrix}$$

A - симметр.

$$\chi_A = \det(A - \lambda I) = \begin{pmatrix} 4-\lambda & & & \\ -4 & 5-\lambda & & \\ 2 & 1 & 14-\lambda & \\ -2 & -2 & -11 & 19-\lambda \end{pmatrix} =$$

$$= (14-\lambda) M_{11} + 4 M_{12} + 2 M_{13} + 2 M_{14}$$

$$M_{11} = (5-\lambda)\left((14-\lambda)(19-\lambda) - 121\right) - (19-\lambda-22) - 2 \cdot$$

$$\cdot \left(-11 + 2(14-\lambda)\right) = -\lambda^3 + 38\lambda^2 - 305\lambda + 694$$

Made with Goodnotes

$$M_{12} = -4(\lambda^2 - 33\lambda + 145) - (2(19-\lambda) - 22)$$
$$-2(-22 + 2(14-\lambda)) = -4\lambda^2 + 138\lambda - 608$$

$$M_{13} = -4(19-\lambda-22) - (5-\lambda)(2(19-\lambda)-22) -$$
$$- 2(-4+2) = -2\lambda^2 + 30\lambda - 64$$

$$M_{14} = -4(-11 + 2(14-\lambda)) - (5-\lambda)(-22 + 2(14-\lambda)) +$$
$$+ (-4+2) = -2\lambda^2 + 24\lambda - 100$$

$$\chi_A(\lambda) = \lambda^4 - 42\lambda^3 + 433\lambda^2 - 1254\lambda + 16$$

λ	0	1	5	6	10	20	30
$\chi_A(\lambda)$	16	-846	-54	304	-1224	-27864	28096

Знаки меняются: (0,1) (5,6) (6,10) (20,30)

$\Rightarrow$ Там есть корни и все корни $\lambda_i > 0$

$\Rightarrow \exists$ Разлож. Холецкого

$$A = \begin{pmatrix} 4 & -4 & 2 & -2 \\ -4 & 5 & 1 & -2 \\ 2 & 1 & 14 & -11 \\ -2 & -2 & -11 & 19 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 & 0 \\ b & c & 0 & 0 \\ d & E & f & 0 \\ g & h & k & \ell \end{pmatrix} \begin{pmatrix} a & b & d & g \\ 0 & c & E & h \\ 0 & 0 & f & k \\ 0 & 0 & 0 & \ell \end{pmatrix}$$

Made with Goodnotes

$a^2 = 4 \Rightarrow a = 2$

$ab = -4 \Rightarrow b = -2$

$ad = 2 \Rightarrow d = 1$

$ag = -2 \Rightarrow g = -1$

$b^2 + c^2 = 5 = 4 + c^2 \Rightarrow c = 1$

$bd + ce = 1 = -2 + e \Rightarrow e = 3$

$bg + ch = -2 = -2 \cdot -1 + 1 \cdot h \Rightarrow h = -4$

$d^2 + e^2 + f^2 = 1 + 9 + f^2 = 14 \Rightarrow f = 2$

$dg + eh + fk = -11 = -13 + 2k \Rightarrow k = 1$

$g^2 + h^2 + k^2 + \ell^2 = 19 = 1 + 16 + 1 + \ell^2 \Rightarrow \ell = 2$

$$L = \begin{pmatrix} 2 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 \\ -1 & -4 & 1 & 2 \end{pmatrix} \Rightarrow A = LL^T$$

Задача № 3

$$A = \begin{pmatrix} 2 & -2/3 \\ 1 & -5/3 \\ 0 & 4/3 \end{pmatrix}$$

$$A = \underset{2\times3}{U}\ \underset{3\times3}{\Sigma}\ \underset{3\times2}{V^T}$$

Made with Goodnotes

$$A^TA = \begin{pmatrix} 2 & 1 & 0 \\ -\frac{2}{3} & -\frac{5}{3} & \frac{4}{3} \end{pmatrix} \begin{pmatrix} 2 & -\frac{2}{3} \\ 1 & -\frac{5}{3} \\ 0 & \frac{4}{3} \end{pmatrix} = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}$$

$$\det(A^TA - \lambda I) = (5-\lambda)^2 - 9 = (\lambda - 8)(\lambda - 2) = 0$$

$$\Rightarrow \sigma_1 = \sqrt{8} \qquad \sigma_2 = \sqrt{2}$$

2) $\lambda_1 = 8$ $\quad (A^TA - 8I) = \begin{pmatrix} -3 & -3 \\ -3 & -3 \end{pmatrix}$

$$\begin{pmatrix} -3 & -3 \\ -3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{matrix} -3x - 3y = 0 \\ -3x - 3y = 0 \end{matrix}$$

v_1

$$\Rightarrow v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\lambda_2 = 2$ $\quad \begin{pmatrix} 3 & -3 \\ -3 & 3 \end{pmatrix} v_2 = 0 \Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

v_1 и v_2 уже ортонорм. $\Rightarrow$

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$u_i = \frac{A v_i}{\sigma_i}$$

$$A = U \Sigma V^T$$

$$AV = U\Sigma$$

$$U = AV\Sigma^{-1}$$

$$u_1 = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 2 + \frac{2}{3} \\ 1 + \frac{5}{3} \\ -\frac{4}{3} \end{pmatrix} = \begin{pmatrix} 2/3 \\ 2/3 \\ -1/3 \end{pmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 2 - \frac{2}{3} \\ 1 - \frac{5}{3} \\ \frac{4}{3} \end{pmatrix} = \begin{pmatrix} 2/3 \\ -1/3 \\ 2/3 \end{pmatrix}$$

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$$A = \underbrace{\frac{1}{3}\begin{pmatrix} 2 & 2 \\ 2 & -1 \\ -1 & 2 \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}_{V^T}$$

Задача №4

$$A = \begin{pmatrix} 0 & 0 & 5 \\ 2 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix} \qquad A = QR$$

Q — ортог., R — верхнее Δ

$$a_1 = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \quad a_2 = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \quad a_3 = \begin{pmatrix} 5 \\ 1 \\ 4 \end{pmatrix}$$

$$q_1 = \frac{a_1}{\|a_1\|} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \|v_1\| = 2$$

$$v_2 = a_2 - (\underbrace{q_1^T a_2}_{3}) q_1 = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$\|v_2\| = 2 \Rightarrow q_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$v_3 = a_3 - (\underbrace{q_1^T a_3}_{1}) q_1 - (\underbrace{q_2^T a_3}_{4}) q_2 = \begin{pmatrix} 5 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - 4\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix}$$

$$\|v_3\| = 5 \qquad q_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

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$$Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \qquad R = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix}$$

Задача №3

1) $A^T A v = \lambda v \qquad v \neq 0 \quad \lambda \neq 0$

$A A^T (Av) = \lambda Av$

$(Av) \neq 0 \Rightarrow A^T Av = 0 = \lambda v$ – противор.

Значит Av – собств вектор AA^T с тем же λ

$AA^T u = \lambda u \qquad u \neq 0 \qquad \lambda \neq 0$

$A^T A A^T u = \lambda A^T u$

$A^T u \neq 0 \Rightarrow \lambda$ – собств ч. $A^T A$ #

2) $A^T A$ – симметр и $\geq 0 \Rightarrow$

$\Rightarrow \exists$ ортонорм. $v_1 \dots v_n$: $A^T A v_i = \lambda v_i$

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$

$x = \sum_{i=1}^{n} c_i v_i \quad ; \quad \sum c_i^2 = 1$

Размышления

$\|Ax\|_2^2 = x^T A^T A x \Rightarrow x^T \cdot (c_i \underbrace{A^T A v_i}_{\lambda v_i}) \Rightarrow x^T \sum c_i \lambda_i v_i$

$\Rightarrow \sum c_i^2 \lambda_i \leq \lambda_1 \sum c_i^2 = \lambda_1$

Made with GoodNotes

Покажем, что:

$$\|Ax\|_2 \le \sqrt{\lambda_1} = \sigma_1$$

Т $x = v_1$

$$\|Av_1\|_2^2 = v_1^T A^T A v_1 = \lambda_1$$

$$\|A\|_2 = \sup_{\|x\|_2 = 1} \|Ax\|_2 = \sqrt{\lambda} = \sigma_1$$ ##